Middlesex University London

Science and Technology

Department of Computer Science

Programming for Data Communication and Networks

CST1510

Module Leader: Anna Wieclaw

2024/25

Duration of the module: 12 weeks

Document Version: 1

Online location of handbook

This handbook can also be accessed via My Learning at:

https://mdx.mrooms.net/mod/resource/view.php?id=3022336

Other formats available

This handbook is available in a large print format. If you would like a large print copy or have other requirements for the handbook, please contact the Disability Support Service disability@mdx.ac.uk

Disclaimer

The material in this handbook is as accurate as possible at the date of production. You will be notified of any minor changes promptly. If there are any major changes to the module you will be consulted prior to the changes being confirmed. Please check the version number on the front page of this handbook to ensure that you are using the most accurate information.

Other documents

Your module handbook should be read and used alongside your programme handbook and the information available to all students on My Learning and MyMDX including the Academic Regulations. Your programme handbook can be found on the My Learning programme page for your programme.

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1. Welcome

This module introduces the fundamental concepts of programming through the use of Python. Recognized as a high-level programming language, Python has rapidly emerged as the leading language in recent years, celebrated for its simplicity and powerful efficiency, particularly in areas such as Data Science and Machine Learning, among other applications. This module offers a comprehensive yet straightforward approach to learning programming, focusing on automating data processes, conducting data analyses, and facilitating communication over networks. It is specifically designed for students with no previous programming experience.

2. The module teaching team

Please see below details of the teaching team for this module.

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Santhosh Menon (Dubai 🇦🇪)		
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	Telephone number:	📞 04 375 4908
	Office hours:	Tue: 12 pm to 3 pm, Fri: 12 pm to 2 pm
Vincent Ching (Hong Kong 🇭🇰)		
	Email:	Vching8@gmail.com
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3. Communication with the teaching team

Staff will communicate with students through various channels, including email, phone, the My Learning module page, lectures, seminars, and both online and in-person meetings.

You may receive urgent messages from the team, either individually or as a group, regarding the module.

It is essential to check your university email regularly.

All staff have designated office hours during which you can drop in without needing to book an appointment.

If you have any issues, please discuss them first with a member of the module team. You can also provide feedback on the module to the module leader, your student voice leader, your personal tutor, or through the end-of-module evaluation survey.

4. Module overview

Aims:

The main aim is to provide you with the opportunity to learn programming effectively. This takes place in the context of learning to solve a range of problems arising in the context of the processing, storing and transmitting data. You will understand the role of a basic but important range of data structures and types, and their use. Several data structures of interest within the networking and security fields will be considered, for example those to represent RTP packets (and associated operations on these) in order that audio- visual data can be sent using transport protocols such as the UDP protocol. The concept of an algorithm will be introduced early on together with the techniques for program implementation. You will be encouraged and supported in undertaking - both individually and within a small group setting - a variety of problem-solving tasks.

Learning outcomes:

Knowledge

On successful completion of this module, the student will be able to:

1. Articulate the use and meaning of a range of key programming constructs.
2. Select (and design) data structures for given programming purposes.
3. Demonstrate and explain basic algorithms and the use of operations on a variety of data structures; searching and sorting of a list, and basic techniques for transferring data between computer components, and between networked computers.

Skills

On successful completion of this module, the student will be able to:

4. Plan, develop, test and debug programs to solve a range of problems using recognised principles.

5. Explain the operation of a selection of given programs.
6. Produce code to a professional standard, including comments, indentation and other meta-information.
7. Operate effectively to carry out collective programming tasks both individually and as a group member.

Syllabus:

Basic terminology: compilers, interpreters, debugging and testing software, variables, expressions and statements data structures and types, operations, functional, functions; function calls, composition; flow of control/execution; parameters and arguments; stacks, frames and return values; program documentation. Conditionals and recursive definition.

Integers, real numbers, mathematical and logical operators; Boolean expressions; nested conditionals; device, machine and network I/O and operations.

Incremental program development; type checking; destructive and non-destructive assignment; iteration; while loops; break statement; algorithms; strings; for-loops; searching and sorting; global variables.

Objected oriented languages, their main features and approach. Object-oriented approach to programming: classes, methods, inheritance and object creation; files; paths; classes and types; instances as return types; the role and use of prototyping. Introduction to sockets and streams.

Learning and teaching strategy:

The formal and semi-formal concepts needed for programming is addressed in (weekly 3-hour) Labs where key principles and concepts are introduced during the Lab. Problem-solving exercises promulgating the understanding and applicability of these will be undertaken in weekly sessions where students can complete tasks according to the Students' Observation Behaviours (SOBs). Each workshop has a set number of SOBs where students are expected to perform these practical tasks before advancing to the next session. Students can make use of the time given in these sessions to complete the threshold SOBs and the whole week to complete the rest of the SOB activities. These SOBs are set in three different categories, Threshold, Typical and Excellent. Once students satisfy the threshold can move to the next session.

Weekly Workshops/Seminars (2 hours): This session offers an inclusive environment for open discussions, concept explanations, group activities, and reinforcing understanding of security, theories, algorithms, and techniques. Through

presentations, active student participation, and group discussions, the aim is to inspire students to explore deeper into topics of interest, fostering research-oriented engagement. These series of workshops are designed to support the student and to consolidate and develop their understanding of basic principles and concepts of programming. These principles and concepts are applied in a range of practical-based problem-solving activities and such activities will be preceded by addressing a problem of interest and providing general frameworks or methods by means of which the problem can be solved effectively.

Weekly online student-focused workshops (1 hour). These sessions are student-led and can host a range of activities listed above and provide amnesty for students who are falling behind in their SOBs and studies.

Briefly, the delivery of this module will be on-campus, over 12 weeks and have:

- Weekly 2-hour workshops;
- Weekly 3-hour labs; and,
- Weekly drop-in session
- Require additional 228 hours of independent study

Key concept videos and additional learning materials are provided on the 'Virtual Learning Environment' to help students with their study.

5. Learning resources

Essential

J. Hunt, *A beginner's guide to Python 3 programming*, 1st ed. Springer,2019.

Recommended

J. Hunt, *Advanced guide to Python 3 programmin*g, 1st ed. Springer, 2019.

A. Sweigart, *Automate the boring stuff with Python: practical programming for total beginners*, 1st ed. No Starch Press, 2019.

A range of academic papers, technical reports, open access books and online resources, e.g., APIs, will be recommended.

6. Expectations of studying this module

Attendance and Engagement

Engaging with online and on-campus in-person learning and activities is integral to your success. Middlesex University supports you to achieve your full potential through a number of strategies, all of which provide a supportive learning environment online, remotely, face-to-face, or blended.
Further information on attendance and engaging with your programme will be available at your Induction and general information is available at the weblink below.
https://mymdx.mdx.ac.uk/campusm/home#pgitem/419149/t

Professional behaviour and online conduct
The programme of study you are undertaking is underpinned by developing professional behaviour and attitude. You are expected to behave in a professional, supportive manner to your peers and teachers. You must come to sessions prepared and ready to contribute where appropriate. Please remember that your University ID should be carried with you always whilst on campus and you must be able to identify yourself if asked to do so. Please conduct your email communication with fellow students, tutors and all relevant staff in a formal and courteous manner.

Academic Integrity and Misconduct
Academic Integrity is a set of principles and values to show that you work in a professional, honest and ethical way. You should be aware of the University's academic integrity and misconduct policies and procedures. Taking unfair advantage over other students in assessment is considered a serious offence by the University. Action will be taken against any student who contravenes the regulations through negligence, foolishness or deliberate intent. Academic misconduct takes several forms, in particular:

- **Plagiarism** – using extensive unacknowledged quotations from, or direct copying of, another person's work and presenting it for assessment as if it were your own effort. This includes the use of third party essay writing services.
- **Collusion** – working together with other students (without the tutor's permission), and presenting similar or identical work for assessment.
- **Infringement of Exam Room Rules** – Communication with another candidate, taking notes to your table in the exam room and/or referring to notes during the examination.
- **Self-Plagiarism** – including any material which is identical or substantially similar to material that has already been submitted by you for another assessment in the University or elsewhere.

Appropriate use of Artificial Intelligence (AI) is detailed in the assessment requirements grid in section 7.5

Student Success Essentials Course includes useful information about how to approach your assessments and complete them with honesty. The course also describes what plagiarism (cheating) is and how to avoid it so you don't face any disciplinary action. For successfully completing this course**, you will be awarded**

Digital-Credentials and a certificate that will verify the knowledge you have gained. Digital-Credentials can be shared and promoted via LinkedIn and other digital channels. You will have to log into to MyMDX and then MyLearning to access the course. https://mdx.mrooms.net/course/view.php?id=17199

Full details on academic integrity and misconduct and the support available can be found at Academic Integrity | MyMDX (mdx.ac.uk)

The Academic Integrity and Misconduct policy is available in our Public Policy Statements (under Academic Quality) at: Our policies | Middlesex University London (mdx.ac.uk)
Referencing & Plagiarism: Suspected of plagiarism?:
http://libguides.mdx.ac.uk/c.php?g=322119&p=2155601
Referencing and avoiding plagiarism:
https://mymdx.mdx.ac.uk/campusm/home#pgitem/419258

The Middlesex University Students' Union (MDXSU) Advice Service offers free and independent support in making an appeal, complaint or responding to any allegations of academic or non-academic misconduct.
https://www.mdxsu.com/advice

Extenuating circumstances:
There may be difficult circumstances in your life that affect your ability to meet an assessment deadline or affect your performance in an assessment. These are known as extenuating circumstances or 'ECs'. Extenuating circumstances are exceptional, seriously adverse and outside of your control. Please see link for further information and guidelines:
https://MyMDX.mdx.ac.uk/your-study/assessment-and-regulations/extenuating-circumstances

7. Assessment

Formative assessment: Formative assessments help show that you are learning and understanding the material covered in this module and allow us to monitor your progress towards achieving the learning outcomes. Although formative assessments do not directly contribute to the overall module mark they do provide an important opportunity to receive feedback on your learning.

Formative assessment	Deadline
Week 1 – 6 SOBs	*Until week 6*

Summative assessment: Summative assessment is used to check the level of learning for the module. It is summative because it is based on accumulated learning during the course. It is the summative assessment that determines the grade that you are awarded for the module.

The table below specifies the associated deadlines:

Summative assessment	Weighting	Deadline	Feedback
Portfolio of Practical tasks	*50 %*	*Week 6*	*7 days after submission*
1000-word report + code and testing	*50%*	*Week 12*	*15 days after submission*

To pass this module, you need to pass all assessments with a minimum grade of 16 with a minimum of grade of 18 in each assessment component

Before you submit your work for final grading, please ensure that you have accurately referenced the work. It is your responsibility to check the spelling and grammar, as all written assessments will assess technical proficiency in the English. This means accurate and effective spelling, punctuation and grammar. Details of how it will be assessed will be provided in the marking criteria for each assessment and the University overall approach can be found within the Grade Criteria Guide in the University Regulations https://www.mdx.ac.uk/about-us/policies (scroll to university regulations)

Reasonable adjustments will be made for those students who have a declared disability/specific learning condition which would affect performance in this area.

Reassessment for this module normally takes place in the following way: 2031 Learning Framework Guidance and Support Material (mdx.ac.uk)

Further information is available at

https://MyMDX.mdx.ac.uk/study/assessment

7.2 Feedback on your assignments

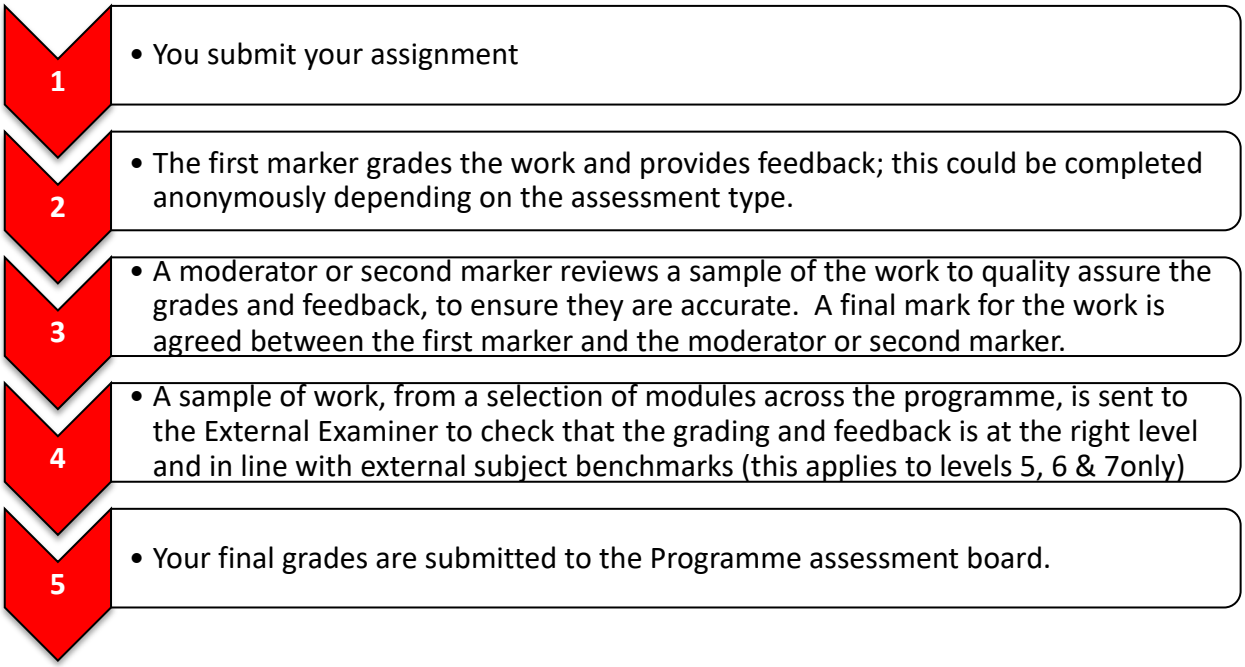
You will be provided with feedback on all assessments that is helpful and informative, consistent with aiding the learning and development process. The nature of the feedback shall be determined at programme level but may take a variety of forms including: written comments; individual and group tutorial feedback; peer feedback; or other forms of effective and efficient feedback. If you have submitted a formative or draft assessment, you will receive feedback but no grade. The comments should inform you about how well you have done or tell you about the areas for improvement. All assignments should be submitted online unless specified in assessment briefs.

Feedback on summative assessments will normally be provided within 15 WORKING DAYS of the published submission date.

7.3 How is your assignment mark agreed?

External Examiners (external academic experts) review what we deliver at a programme level. The University reviews a sample of your work to quality assure the grades and feedback you received from the person who marked your work. Our External Examiners will sample a selection of modules from a programme, with more focus of outcomes between modules within a programme.

The following diagram provides an overview of the marking process for your module assessment. Further information on the role of external examiners can be found at. https://www.mdx.ac.uk/about-us/policies/academic-quality/handbook (section 4)



Results Confirmation

Module grades are considered and confirmed by an Assessment Board. Following the Board, individual results and your progression status or finalist classification will be made available in the My Study area on MyMDX.
If any of your results are provisional they will be labelled as such. Further information on your results and assessment can be found in the University Guide, under the Your Study area of MyMDX, or by seeking advice from your Progression and Support Team Officer. Further details can also be found in the University regulations.

7.4 Anonymous Marking Assessment Policy

We have worked with the Middlesex University Students' Union (MDXSU) to create an anonymous marking policy, in response to student feedback. Anonymous marking ensures that your identity (your name, student number and other personal/identifiable information) is not made available to academics when they are marking your work. This means that you can have confidence that your assessments will be marked fairly and consistently. However, there are some forms of assessment for which anonymity cannot be guaranteed and these are recognised

in the policy. The Anonymous Marking Assessment Policy is available at: https://www.mdx.ac.uk/__data/assets/pdf_file/0037/563599/anonymous-marking-assessment-policy.pdf

We now look at each component of summative assessment for this module in detail. Each of the following tables provides an overview of the requirements.. The support provided for each component along with the feedback arrangements, is also detailed below.

7.5.1 Assessment 1

The following table provides an example of the overview of the assessment requirements.

PORTFOLIO OF PRACTICAL TASKS	
Module code	*CST1510*
Module title	Programming for data communication and networks
Submission date, time	Week 6
Feedback type & date	After 15 working days
Word count	N/A
Assignment type	Portfolio of Practical tasks.
Assignment structure, format and details	A collection of tasks demonstrating understanding and practical application of skills, showcasing your grasp of Pytho fundamentals up to this point including: o Basic Python syntax, handling input/output, using variables. o Using conditionals, loops, and working with lists. o Creating functions, using lambda, working with nested lists. o Advanced string handling, using modules, applying regex. o File I/O, exception handling, using tuples, sets, dictionaries, and data manipulation with Pandas.
Appropriate use of AI	The aim is for you to become the "clockmaker"—understanding how to program and use AI tools wisely. While AI can assist with code review and feedback, it's essential to evaluate its suggestions critically. Use AI to enhance your skills, not replace them. *In your portfolio, specify when and how you used AI tools, crediting their contributions appropriately to ensure transparent and ethical use of technology.*
Assessed learning outcome (s)	3,5,6,7
Assessment weighting %	50%
Key reading and learning resources	Weekly material. Essentia reading

7.5.2 Assessment 2

1000-WORD REPORT + CODE AND TESTING					
Module code	*CST1510*				
Module title	Programming for data communication and networks				
Submission date, time	Week 6				
Feedback type & date	After 15 working days				
Word count	N/A				
Assignment type	Portfolio of Practical tasks.				
Assignment structure and format	A 1000-word report and a Python-developed software project are required. Detailed specifications and a template will be available on the MyLearning page.				
Appropriate use of AI	You are free to use AI for checking grammar, spelling, and structuring your written material but the content value must be your own. All code should be your own work; if you use AI to generate any code, specify which parts. While AI can suggest approaches, you should write the code yourself. If your work appears fully generated by AI, you may be required to attend an additional viva examination. *In your report and code, clearly state when and how you used AI tools, giving appropriate credit to ensure transparency and ethical use of technology.*				
Assessed learning outcome (s)					
Module weighting %	50%				
Key reading and learning resources					
Criteria	**1-4 First**	**5-8 Upper Second**	**9-12 Lower Second**	**13-16 Third**	**17-20 Refer**
	70%+	**60%-69%**	**50%-59%**	**40%-49%**	**Less than 40%**
Report:	Exceptional depth and clarity in explaining the problem, methodology,	Thorough discussion of the problem, methodology, and outcomes with well-	Adequate explanation of the problem, methodology, and outcomes.	Basic discussion of the problem, methodology, and outcomes. Justification of the	Inadequate explanation of the problem, methodology, and outcomes. Little to

	and outcomes. Strong justification for the approach used, with clear links to relevant theories and programming principles	supported justification of the chosen approach. Shows a strong understanding of programming concepts.	Justification of the approach is present but lacks depth. Basic understanding of programming concepts is shown.	approach is weak or unclear. Minimal understanding of programming concepts is evident.	no justification of the approach. Lacks fundamental understanding of programming concepts.
Software					
Functionality:	Software meets all requirements with advanced features implemented correctly.	Software meets most requirements with additional features correctly implemented.	Software meets basic requirements with limited additional features. Code is functional but may have some inefficiencies.	Software meets minimum requirements but lacks additional features. Code is often inefficient, poorly structured.	Software does not meet minimum requirements. Little to no evidence of coherent structure or documentation.
Code Quality:	Demonstrates exceptional understanding of Python	Good use of programming constructs. Shows a strong grasp of Python concepts with good structure and readability.	Satisfactory use of basic programming constructs. Code shows a reasonable understanding.	Limited use of programming constructs with numerous errors. Demonstrates a basic or superficial understanding.	Poor use of programming constructs with frequent errors. Shows little understanding of basic programming

8. Learning Planner

- ***Week 1: Getting Started with Python***
  - ***Topics Covered:*** *Variables, simple data types (int, float, string), basic I/O operations (input, print), and introductory Python syntax.*
  - ***What You'll Learn:*** *This week, you'll get comfortable with the basics of Python programming. We'll set the stage for your journey into more advanced topics by making sure you understand the fundamental building blocks of Python code.*
- ***Week 2: Control Structures and Lists***
  - ***Topics Covered:*** *Conditional statements (if, elif, else), loops (for, while), basic list operations, and list comprehensions.*
  - ***What You'll Learn:*** *You will learn how to control the flow of your programs and handle lists, making your code smarter and more efficient. By the end of this week, you'll be able to write programs that can make decisions and work with collections of data.*
- ***Week 3: Functions, Lambda Expressions, and Multidimensional Lists***
  - ***Topics Covered:*** *Defining and using functions, lambda expressions for concise coding, and working with multidimensional lists (nested lists).*
  - ***What You'll Learn:*** *This week, you'll dive into creating reusable code with functions and explore more advanced data structures. You'll also learn how to write more concise and powerful functions using lambda expressions.*
- ***Week 4: Strings, Special Methods, Modules, and Regular Expressions***
  - ***Topics Covered:*** *Advanced string manipulation, special methods, using modules and packages, and an introduction to regular expressions.*
  - ***What You'll Learn:*** *While learning about strings and regular expressions, you'll discover powerful ways to process and manipulate text data. We'll also explore Python's modularity, showing you how to leverage existing code to build more complex applications*
- ***Week 5: File Handling, Exceptions, and Data Structures including Pandas***
  - ***Topics Covered:*** *File I/O (reading, writing, and parsing), handling exceptions, a deep dive into tuples, sets, and dictionaries, and an introduction to Pandas for data manipulation.*
  - ***What You'll Learn:*** *You will gain the skills needed to work with files and manage data robustly. You'll also get a taste of data structures, including Pandas, a powerful tool for manipulating large datasets.*
- ***Week 6: Portfolio Submission and Project Work***

- ***Topics Covered:****Portfolio compilation of work from Weeks 1-5, covering variables, control structures, functions, file handling, exception management, and Pandas.*
- *Start on a comprehensive project integrating all learned skills.*
- ***What You'll Learn:****You'll showcase your understanding through a portfolio of key assignments, reflecting on your progress so far. You'll also begin a practical project that ties together all concepts from the first half of the course, applying your skills to a real-world problem.*

- ***Week 7: Object-Oriented Programming, Inheritance, and Polymorphism***
  - ***Topics Covered:*** *Core OOP principles (classes, objects), inheritance, polymorphism, and applying these concepts in real-world scenarios.*
  - ***What You'll Learn:*** *This week, you'll dive into object-oriented programming, which is essential for writing complex, organized, and reusable code. You'll learn how to model real-world problems using classes and objects.*
- ***Week 8: Networking and Data Analysis***
  - ***Topics Covered:*** *Advanced networking concepts using Python (socket programming, protocols like HTTP/HTTPS) and data analysis techniques focused on networking data.*
  - ***What You'll Learn:*** *You'll integrate your networking skills with data analysis, learning how to build simple networked applications and analyze data.*
- ***Week 9: SQL and NoSQL Databases with Python***
  - ***Topics Covered:*** *Working with SQL databases (sqlite3, SQLAlchemy) and NoSQL databases (e.g., MongoDB) and their integration with Python.*
  - ***What You'll Learn:*** *This week, you'll master the skills needed to interact with both relational and non-relational databases. You'll learn how to store, retrieve, and manipulate data efficiently in Python*
- ***Week 10: GUI Development with Tkinter***
  - ***Topics Covered:*** *Building graphical user interfaces (GUIs) using Tkinter, including layout management, widgets, event handling, and creating interactive applications.*
  - ***What You'll Learn:*** *You'll learn how to create user-friendly interfaces for your Python applications using Tkinter. By the end of this week, you'll be able to design and implement basic GUIs, enhancing the usability of your software projects.*
- ***Week 11: Web Scraping and Working with URLs and Requests***

- ***Topics Covered:*** *Web scraping with libraries like BeautifulSoup and Scrapy, and handling HTTP requests with the requests library.*
- ***What You'll Learn:*** *You'll learn how to gather data from the web, a valuable skill in data communications. By the end of this week, you'll be able to extract and process information from websites.*

- ***Week 12: Final Project***
  - ***Assessment:*** *A comprehensive project that brings together all the skills and knowledge you've gained throughout the course.*
  - ***What to Expect:*** *In your final project, you'll have the opportunity to demonstrate your mastery of the entire course. You'll tackle a real-world problem involving programming, data communications, and networking, showcasing your ability to apply what you've learned in a practical, integrative way.*

9. University 20-point Scale

20-point scale	General scale	General scale (full ranges)	Percentage used for aggregation purposes only (for areas marking directly to the 20 point scale on modules with multiple assessment components)
1	80% - 100%	79.50% - 100%	90%
2	76% - 79%	75.50% - 79.49%	77.5%
3	73% - 75%	72.50% - 75.49%	74%
4	70% - 72%	69.50% - 72.49%	71%
5	67% - 69%	66.50% - 69.49%	68%
6	65% - 66%	64.50% - 66.49%	65.5%
7	62% - 64%	61.50% - 64.49%	63%
8	60% - 61%	59.50% - 61.49%	60.5%
9	57% - 59%	56.50% - 59.49%	58%
10	55% - 56%	54.50% - 56.49%	55.5%
11	52% - 54%	51.50% - 54.49%	53%
12	50% - 51%	49.50% - 51.49%	50.5%
13	47% - 49%	46.50% - 49.49%	48%
14	45% - 46%	44.50% - 46.49%	45.5%
15	42% - 44%	41.50% - 44.49%	43%
16	40% - 41%	39.50% - 41.49%	40.5%
17	35% - 39%	34.50% - 39.49%	37%
18	30% - 34%	29.50% - 34.49%	32%
19	0% - 29%	0.01% - 29.49%	15%
20	Non-participation	0%	0% (non-submission of a component)